Correction of Article Errors in Machine Translation Using Web-based Model

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Abstract-**In this paper, an approach is proposed for correcting article errors in English translation results in order to improve the performance of a MT system. We check the article and the singular/plural form of the headword in a NP at the same time. This is different from most of early researches in which only articles are considered. Our correcting algorithm is based on simple, viable n-gram model whose parameters can be obtained using the WWW search engine Google. Using much less features than those used in the early researches, we experimentally showed that our approach could perform the promising results with a precision of 86.2% on all classes of article errors.**

I. INTRODUCTION

Recently, Machine Translation (MT) has made significant achievements. However its quality has been known to be very poor, and has not reached users' satisfactory level. With the rapid progress of Internet technology, international conversation becomes more frequent. The necessary of improvement of its performance proves to be even more serious. In [1], Schafer established a special error typology after gathering examples of errors typically corrected by post-editors in their various MT projects. In their classification, the translation error classification includes lexical errors (e.g., terminology and idioms), syntactic errors (e.g., word order and clause analysis) and grammatical errors (e.g., article and tense).

In this paper, we address on the problem how to automatically correct article selection errors and the singular/plural form errors of the headword modified by this article in order to improve the performance of MT systems. The reason that we process the two kinds of errors together is that how to use an article is largely depended on the singular/plural form of the headword in this phrase. It certainly seems to be more reasonable for us to check article and the singular/plural form of the headword together. For example, for the translation result "#[1]It could be a growing pain", our research goal is to automatically correct the phrase "#a growing pain" to "growing pains", for *growing pains* is a plural only compound noun which always occurs in plural form and there should be no article in this sentence either. In the following discussion, we call these kinds of errors as "article errors" which include both article selection errors and singular/plural form errors.

[1] "#" means that the text is grammatically wrong.

The article selection is to decide when to use a (an), the, or zero articles at the beginning of a noun phrase (NP), in which the singular/plural form of the headword also need to be determined accordingly. It is one of the most complex problems in translation result generation in MT. Especially when source texts are written in some languages, such as Chinese and Japanese, which do not have any articles or mark the countability, the problem becomes more difficult.

In this paper, after some discussion of the related research (section 2), we introduce an approach to correct article errors. In our correction algorithm, simple and viable n-gram-based model is proposed to select the correct target NP from candidates. The parameters of the model (web counts of queries) can be obtained with the help of WWW search engine Google (section 3). We evaluated the correction ability of our approach for all article error classes on the revised test set. In order to learn the performance on each error class separately, we made up several special test sets using artificial errors and corrected sentences to realize the evaluation. We show that our algorithm performs the promising results (section 4). Finally the paper ends with some conclusions and future work (section 5).

II. RELATED RESEARCH

Most of the early researches on article problem have been using rules to improve the quality of text [2, 3]. In the research of Japanese and English translation, Heine [3] focused on Japanese NP and classified whether it is definite or indefinite. The detection rules were all extracted by hands. The result that 79.5% of the NPs were classified with an accuracy of 98.9% was reported.

However writing a complete article rule set is a time-consuming work which also needs the help of linguists. Contrasting to these hand-made rules, Knight and Chander in their work [4] proposed an automatic post-editor to insert articles into English. They looked article selection as a classification problem, and trained a decision-tree builder on 4,000,000 NPs from *Wall Street Journal* text to learn whether to generate *the* or *a/an*. They used a variety of lexical (e.g., words before or after the article), syntactic (e.g., POS), and semantic (e.g., tense) features to generate over 200,000 rules automatically. They achieved an overall accuracy of 78%.

The cases of zero article that Knight and Chander did not consider was further extended by Minnen et al. in their research [5]. Some additional features were added such as the head of the NP, the presence of a determiner in the NP and

match. *frequency(context)* is the frequency of a certain context, which is estimated by Google "* query" using "*" operator. "*" can stand for any word in a search term. Google search also supports the Boolean operator. To retrieve pages that include either word A or word B, use an uppercase OR between terms. In our searching, we introduce this operator "OR" to expand a search term into all its morphological forms for more accurate estimation.

For example "She cut the apple in two", when we would like to estimated the frequency of "the apple" given the context *(1,1)*, we query "cut the apple in" using Google and obtain its frequency of 541. The frequency of the context is 2,460,000 estimated by the query "cut * * in", in which " * * " stands for any two possible words. We also can apply "OR" to expand "cut" for two other morphological forms: "cutting" and "cuts". Using "(cut OR cutting OR cuts)" instead of "cut" for queries in some cases, we can get more accurate estimation.

Querying in WWW adds noise to the data, we certainly lose some precision compared to supervised statistical models, but we assume that the size of the WWW will compensate the rough queries. Keller and Lapata [9] showed the evidence of the reliability of the web counts for natural language processing. Although they also experimentally showed that web-based approach could overcome data sparseness for bi-grams in [10], the problem still exists in our experiments. When the web count returned is zero, we smooth zero by adding it to 0.01.

Although we used Google as our search engine, we did not use Google Web API service for programme realization in our later experiments, for Google limits to 1000 automated queries per day. As we just need web counts returned for each search query, we directly extracted these numbers from the web pages found.

D. *Processing Flow*

Fig. 1 gives an overview of our processing flow for correcting the article errors.

For a translated English sentence, we firstly need to extract the NP in it (e.g., "a student" in "I am a student") to process in the further using. We use Apple Pie Parser [11], which is a bottom-up probabilistic chart parser. It finds the parse tree with the best score by best-first search algorithm and has been said to perform well for noun phrases especially.

The next step, we classify the extracted NP, and generate the other article candidates. For example, "a student" belongs to C_1, and then we generate "the student" (C_2), "student" (C_3), "the students" (C_4) and "students" (C_5) as article candidates.

We pluralize nouns based on general rules. For example, words that end with *-ch, x, s* or *s* require an *-es* for the plural. Compound nouns create some problems when we need to pluralize them. Since no real rules exist for how to pluralize all the words, we summarized from "*Guide to English Grammar and Writing*" [12] and processed our experimental data following the rules below.

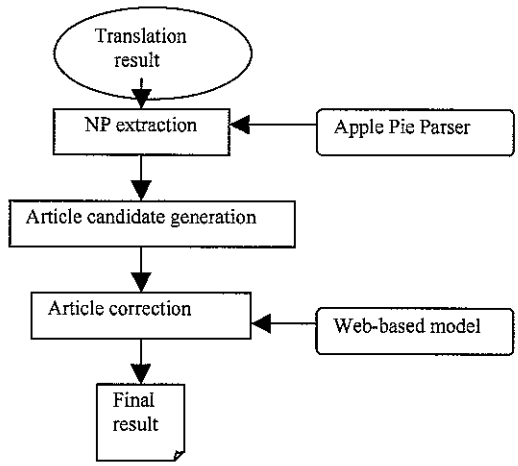


Fig.1. Flow of Article Correction Processing

1. Pluralize the last word of a compound noun (e.g., bedrooms, film stars).
2. When "*woman*" or "*man*" are the modifiers in a compound noun, pluralize both of the words (e.g., Women-drivers).
3. When a compound noun is made up as "noun + preposition (or prep. phrase)", pluralize the noun (e.g., fathers-in-law).
4. When the compound noun is made up as "verb (or past participle) + adverb", pluralize the last word (e.g., grown-ups, stand-bys).

Then using the web-based model explained in section 3.3, we can get final corrected result that is with largest probability estimated by this model.

E. *An Example*

For a better understanding of our approach, we give an example to show how it works. Suppose that we have the following translation sentence that we want to correct article error if any. We assume that there are no other errors in the sentence.

"I have consented on the conditions that he should pay beforehand."

Using Apple Pie Parser, we can extract the NP " the conditions". Then we generate the other article classes, "a condition", "the condition", "condition", "conditions" for article candidates.

Suppose the context is *(2, 1)*. For "a condition", "the condition" and "the conditions", these NPs have 2 words, we generate the query "(consented OR consent OR consents) on * * that" for them while use "(consented OR consent OR

TABLE V

CORRENTION ACCURACY FOR ERROR CLASSES

Class	Recall (%)	Precision (%)	F-score (%)
- a/an	87.7	88.9	88.3
- the	88.7	90.4	89.5
- s	82.5	86.7	84.5
+ a/an	81.1	85.8	83.4
+ the	78.8	82.3	80.5
+ s	79.1	89.2	83.8

total article errors). The correcting precisions of "-a/an", "-the" and "-s" are 88.9%, 90.4% and 86.7% respectively, and the total precision of *loss* class is 88.7% while the precision of *unwanted* class is 85.8%.

From the results we can find out that our approach performs much better in *loss* errors than in *unwanted* errors. This maybe can be explained for some reasons. The context of NPs in which an article is really necessary has much closer relationship with the NPs so that it is more effective to be used in our algorithm. For *unwanted* error class which also occurs frequently, we would like to define some general rules in the further to detect the situations of using an article when there should be none. For example, some countable nouns of institutions are used without articles (e.g., he is in church /college /jail /class), but the rules could be more complex than that.

V. CONCLUSION

In this paper, we explained how to correct article errors in English translation outputs in order to improve the performance of MT systems. Different from the early researches in which only article is considered while other parts are assumed to be correct, we check the article and the singular/plural form of the headword in a NP together. We proposed an unsupervised web-based model whose parameters can be obtained using the WWW search engine Google. We experimentally showed that our approach based on this simple model could perform the promising results with a precision of 86.2% on all article error classes, and 88.7% on *loss* error class.

As part of our future work, we would also like to combine some general rules with our approach to improve the correct rate.

Using WWW is an exciting direction for NLP, but the web-based methods invariably introduce noise in the resulting frequency data. How to eliminate noise data is the key to improve web-based methods. Our next step is aiming at evaluating the Internet resource by distinguishing the useful and noise data.

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